SiPM *Sous-vide:* Testing Temperature-Dependence of Dark Current in Cosmic Watch Silicon Photomultipliers

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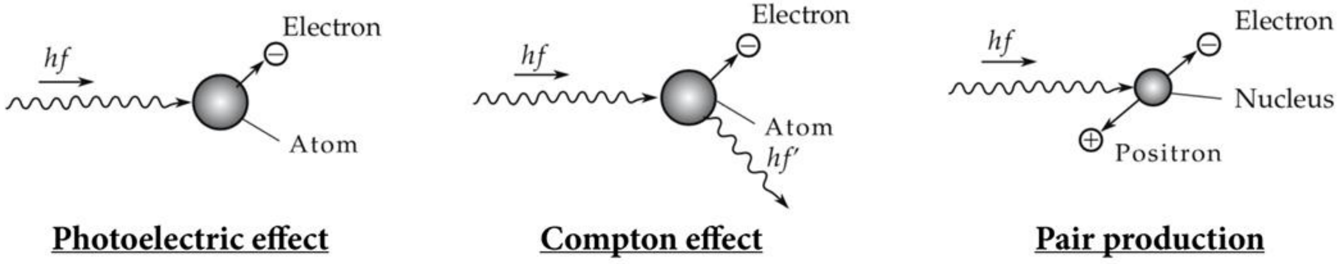
Detecting Single-Particle Events

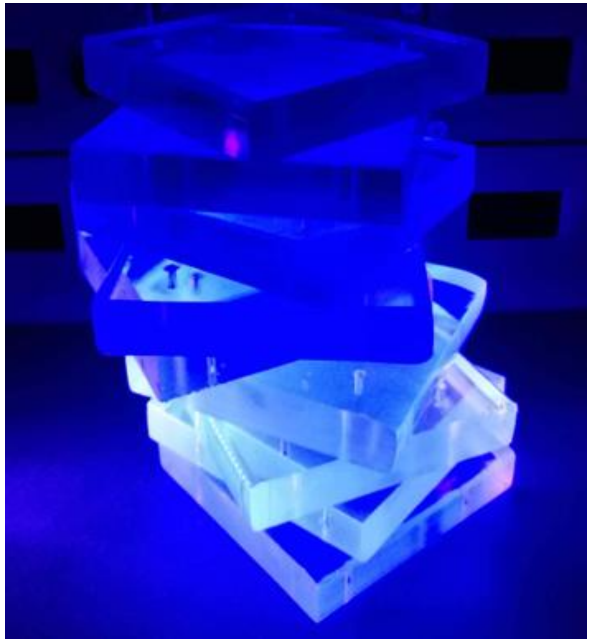
- **Our atmosphere is full of high-energy particles and waves that interact with matter, known as ionizing radiation:** gamma rays, alpha/beta particles, muons

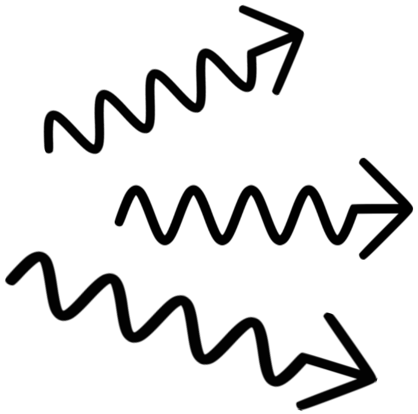


- Measuring the rate and energy of these interactions gives insight into the fundamental nature of ionizing radiation

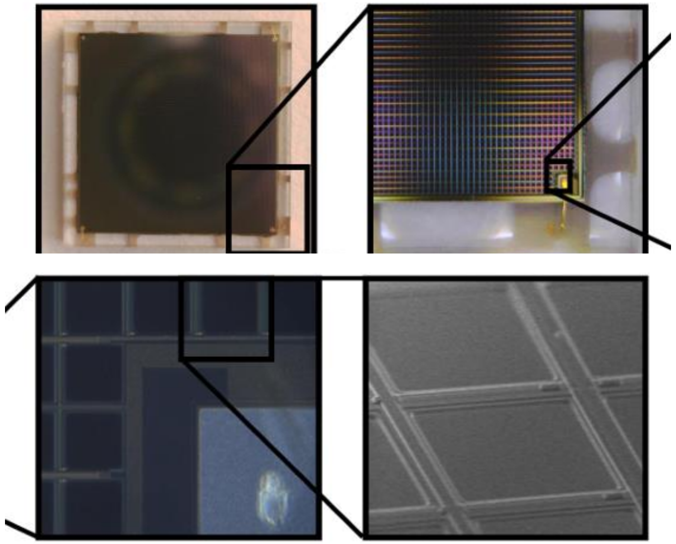
Tools for detecting ionizing radiation

- **Scintillators:** absorbs energy from ionizing radiation -> photons



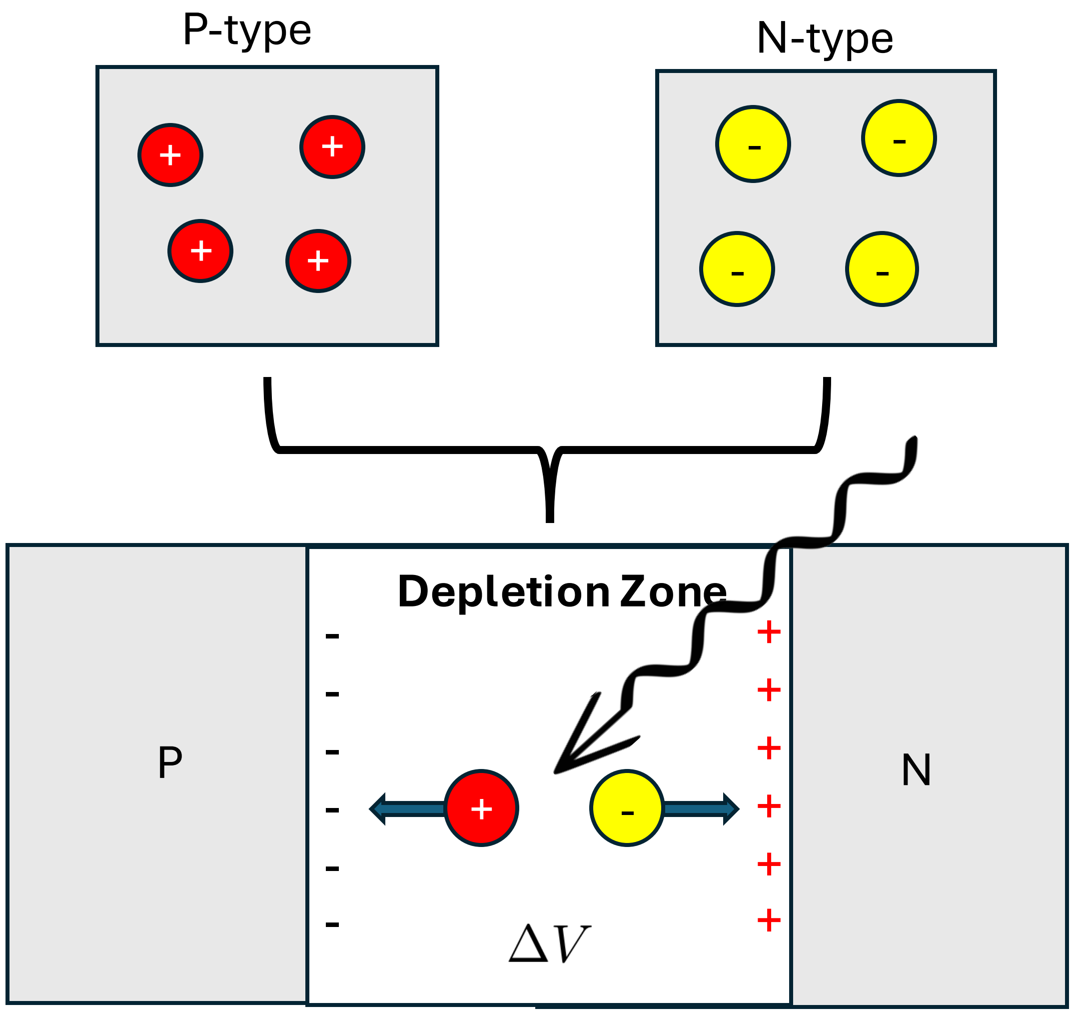


- **SiPM (Silicon Photomultiplier):** reacts to incident photons with "electron avalanche" to produce a measurable current

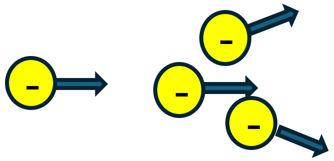


How do we know what we are detecting is really from ionizing radiation? What about "dark current"?

SiPM thermal fluctuations generate electron avalanches like a chemical reaction



- **Activation Energy:** E_a ~ 1.1 eV* (Band-gap of Silicon)
- **Electron Avalanche:** if a PN junction is operated in Geiger mode it triggers an electron avalanche with high probability

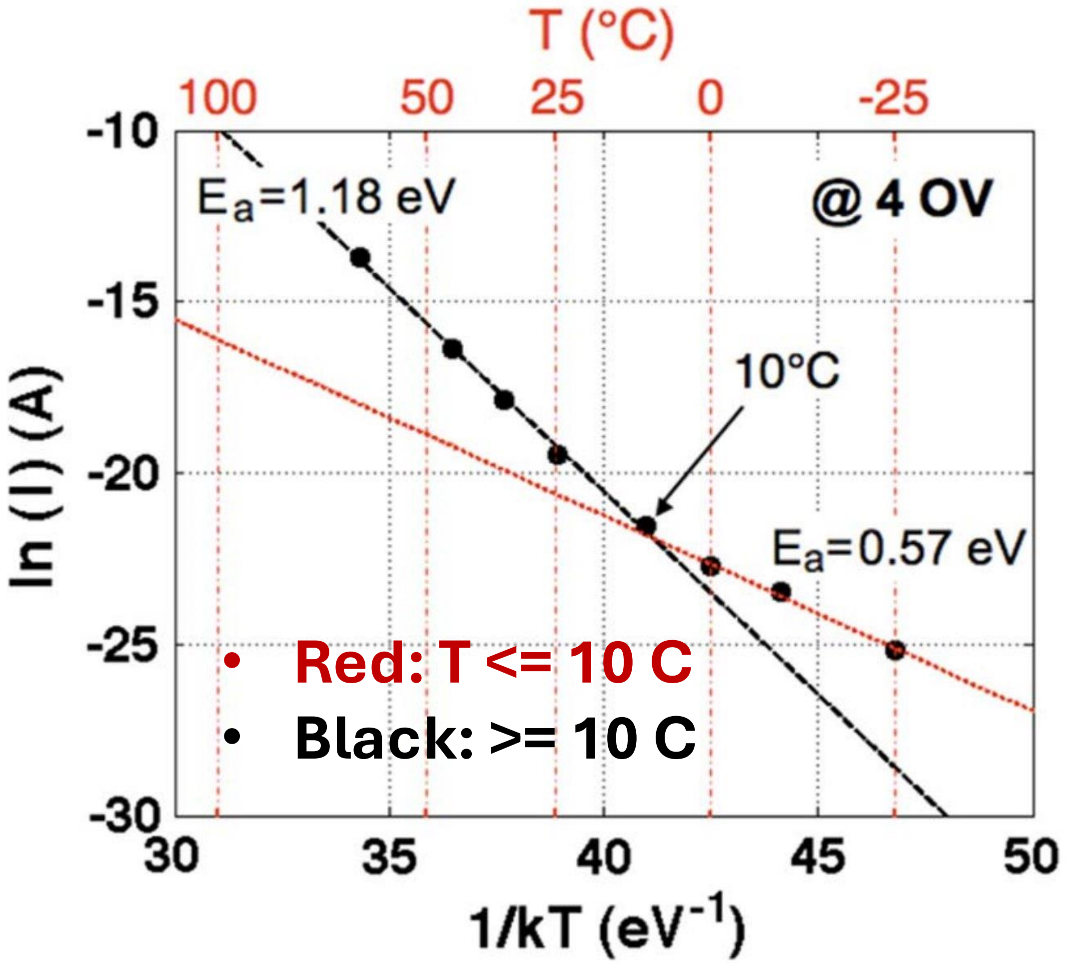


Shockley-Read-Hall Formula:

$$R(T) \propto T^{\gamma} \exp\left(-\frac{E_a}{k_B T}\right)$$

R: Rate (counts/time) of generation due to thermal fluctuations

Previous study shows different temperature regimes of SRH formula



$$\ln R = \gamma \ln T - \frac{E_a}{k_B} \cdot \frac{1}{T} + C$$

- Different temperature regimes have different activation energies, gamma based on charge carrier mechanism (eg. traps) in specific semiconductor

- **Simplest model:** γ=2 (typically 1-3)

Can we observe similar Arrhenius trends in our Cosmic Watch's SiPM by measuring the total count rate at different temperatures T?

Source: Pagano et. Al, https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=6262467

Determination of the event rate R using a Cosmic Watch Detector

Cosmic Watch Measurement Capabilities

- Records single-particle detection events; **elapsed time (ms)**, **temperature (C)**, pressure (Pa), and response voltage (mV), and **dead time (us)** are recorded per event

Quantifying the net event rate R:

$$T_{\text{live}} = T_{\text{elapsed}} - \sum_{i=1}^{N} \delta t_i^{\text{dead}},$$

$$R = \frac{N}{T_{\text{live}}},$$

$$\sigma_R = \frac{\sqrt{N}}{T_{\text{live}}},$$

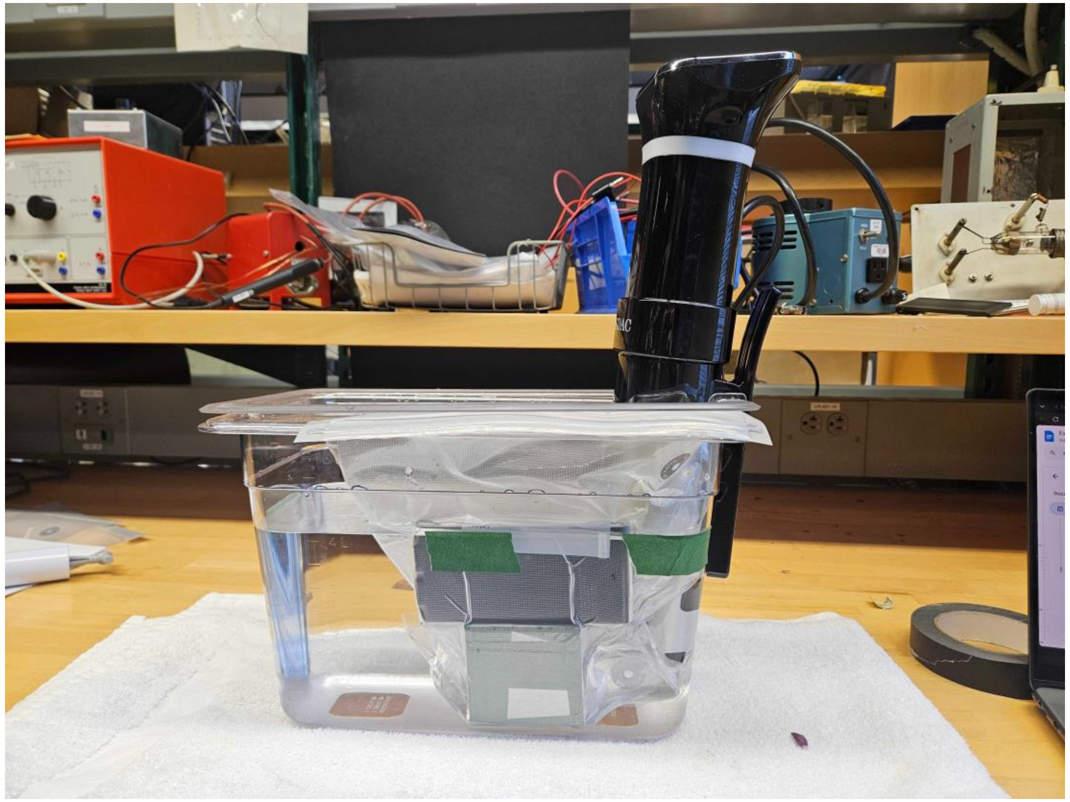
The Ingredients: Using a *sous-vide* bag to heat the Cosmic Watch

- 1 Cosmic Watch
- 1 Plastic air-sealable *sous-vide* bag
- Two brass weights (500g, 100g); anchor CW in water
- 1 Elecom Portable Charger Bank (T_max ~ 60 C)

Each time the CW needs to be turned off/on, this setup must be rearranged, creating systematic orientation-dependent effects.

The Cookware: high and low-temperature apparatus environments for measuring event rate R

High-T Setup: Sous-vide Immersion Cooker



T = 25 – 55 C

J-Lab Staff Fridge (Low T Setup)

T = 3.5 +/- 0.5 C

Both apparatus configurations were placed in close proximity (systematic error: differing background radiation) and contained the same water level

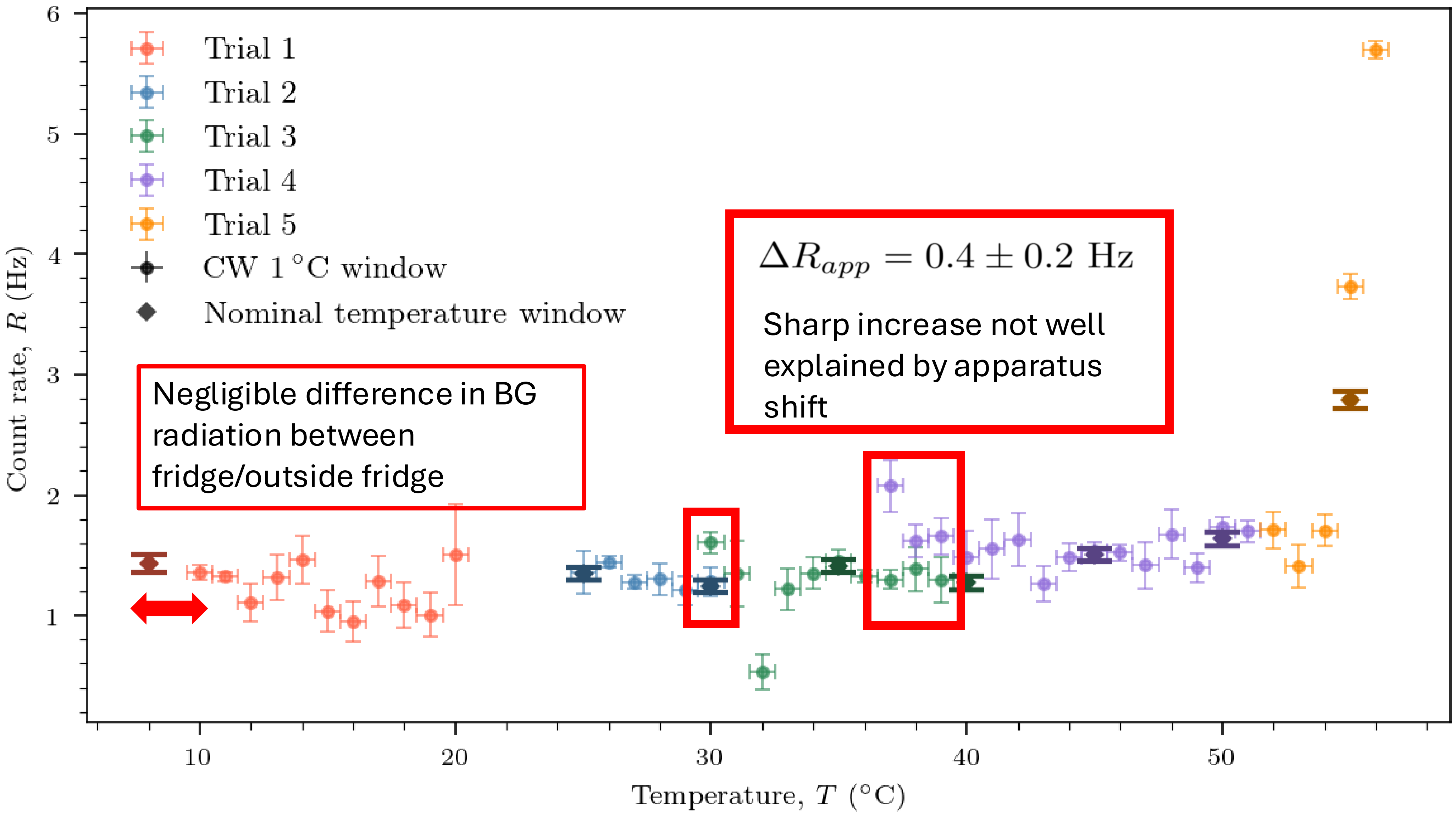
The Recipe: Experimental Procedure for determining T-dependent event rate R

Constraint: CW turns off after 15-20 minutes of being in the water

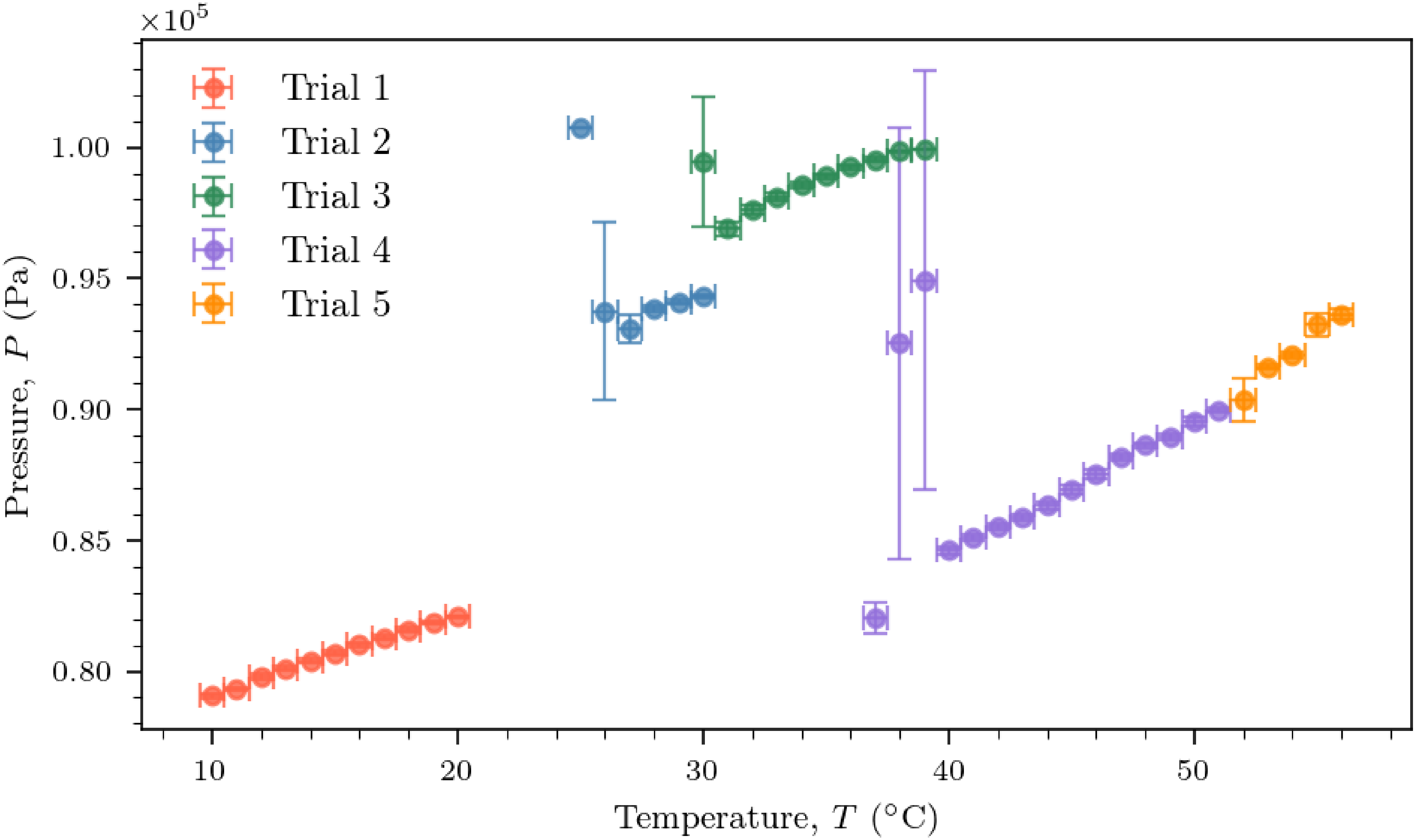
1. **Low Temperature Sequence:** Place CW bag in sous-vide container; leave for approximately 20 hours overnight
   2. Turn on CW and record data for 20 minutes

2. **High Temperature Sequence:** Attach and turn on heater
   1. Using the sous-vide machine display, heat up CW to next closest target (nominal) temperature: 25, 30, 35, 40, 45, 50, 55 C
   2. After target temp is reached, collect data for 9 minutes at constant (nominal) temperature
   3. Repeat until CW turns off, at which point take out CW bag and turn on CW, requiring rearranging the CW orientation within the bag/resealing the bag (systematic effect)

Count Rate R shows sharp increase at T = 55 C



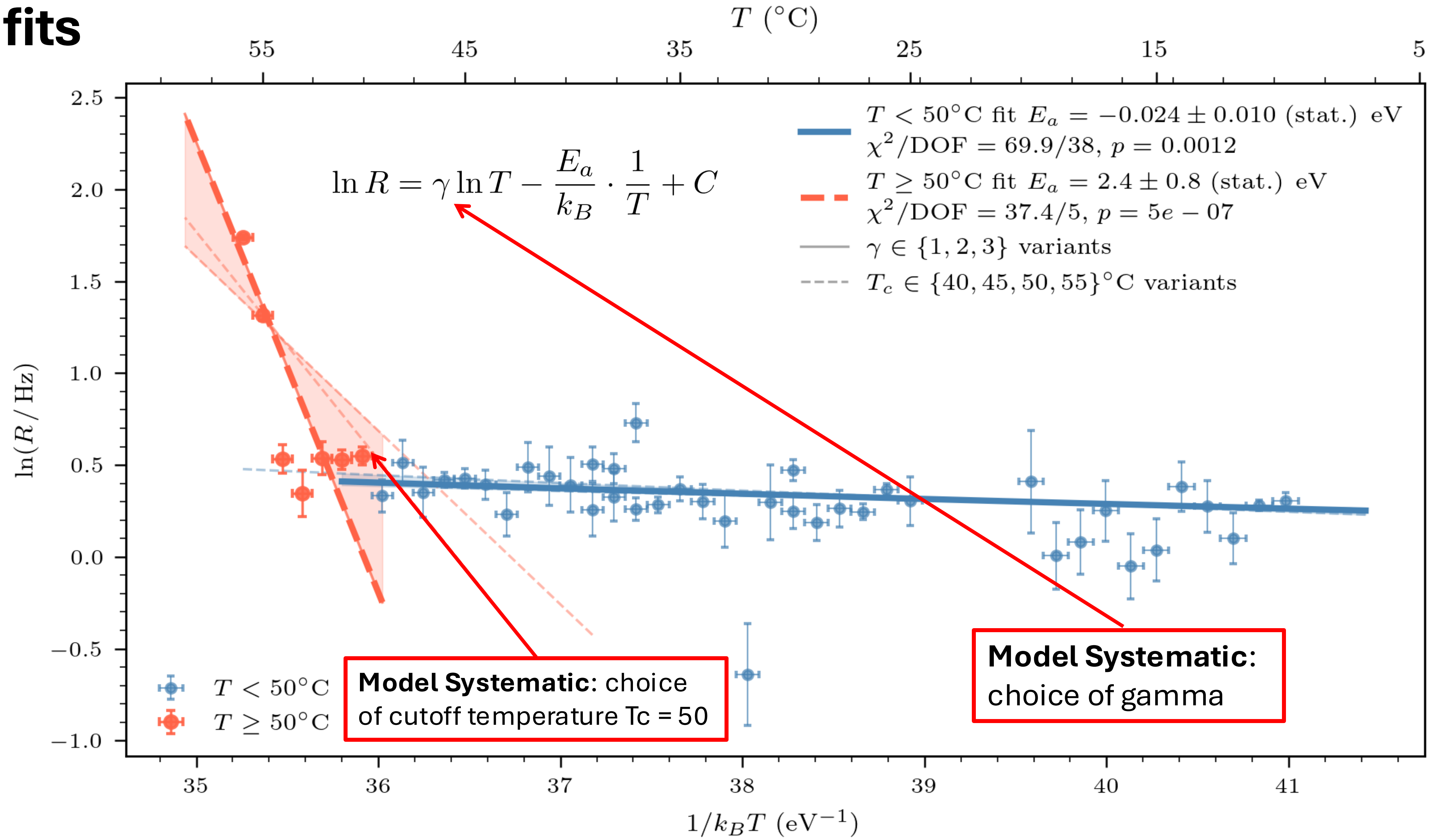
Pressure trend does not explain count rate increase at T = 55 C



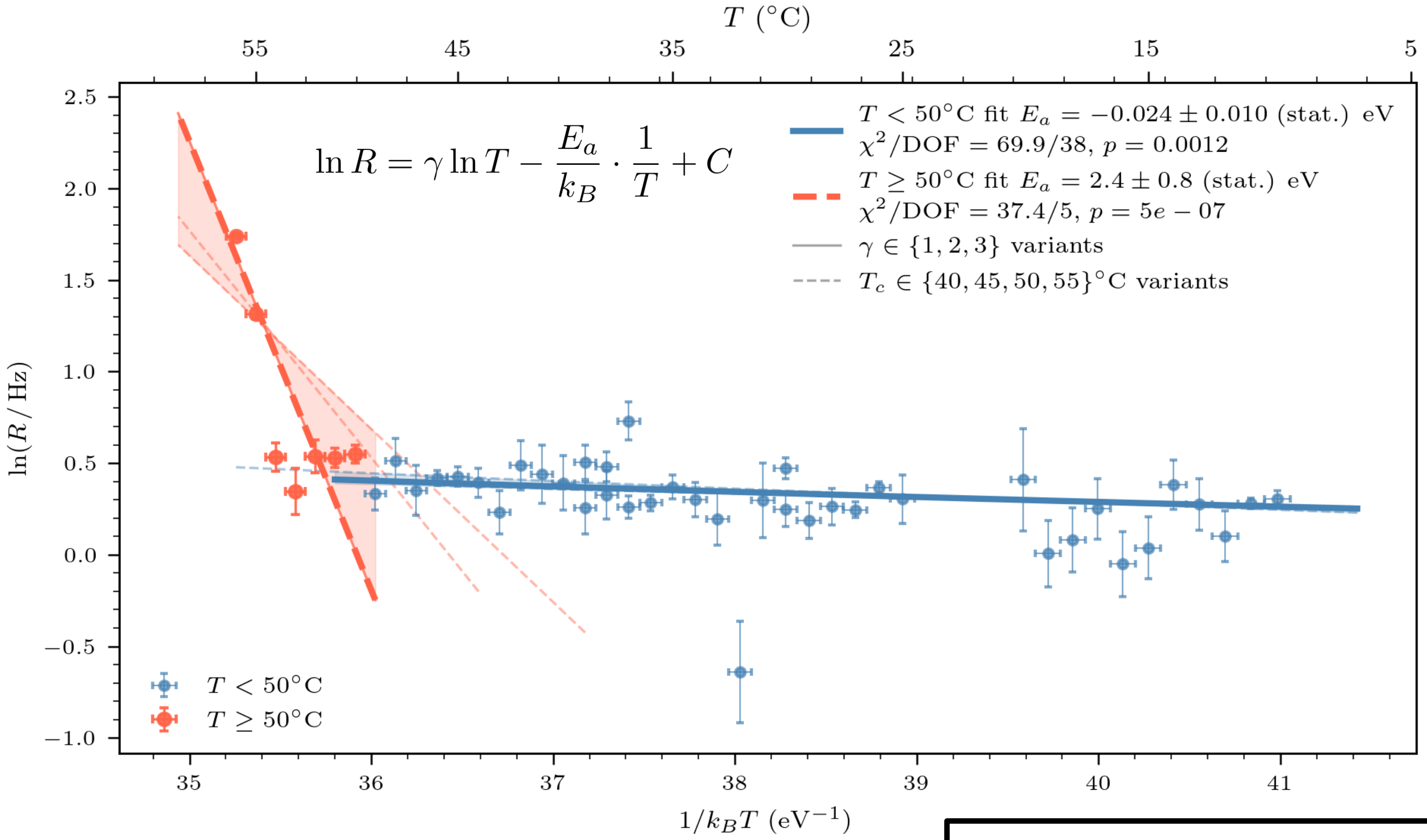
Apparatus arrangement (how air-tight, how much is the CW being compressed) likely contributes a large degree to the measured pressure of the CW

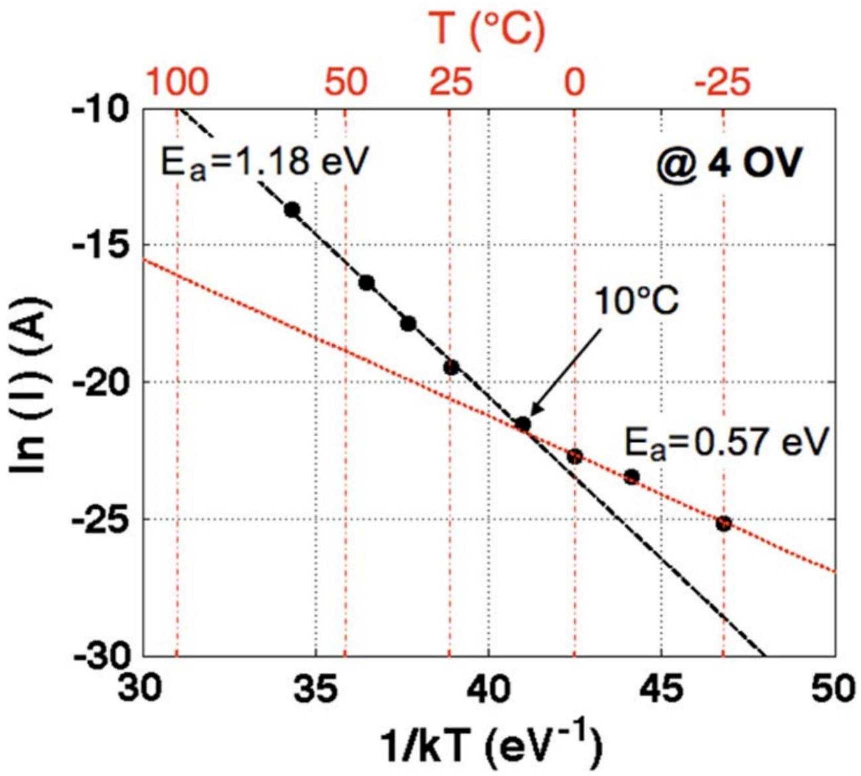
No pattern from T = 50-55 C distinct from lower T's

Arrhenius Plot shows two temperature regimes for SRH model fits



Arrhenius Plot shows two temperature regimes; E_a values do not match previous study





Disagreement in magnitude and temperature of SRH regimes

Calculating Systematic Model Uncertainties

$$E_a^{\text{low}} = -0.024 \pm 0.010\,(\text{stat.})$$

$$E_a^{\text{high}} = 2.4 \pm 0.8\,(\text{stat.})$$

Model Uncertainty in Gamma: how much does varying gamma change the activation energies?

$$\sigma_{\text{sys}}^{(\gamma)} = \frac{|E_a(\gamma = 3) - E_a(\gamma = 1)|}{2}$$

T < T_c: $\sigma_{\text{sys}}^{(\gamma)} = 0.03\ \text{eV}$ **T >= T_c:** $\sigma_{\text{sys}}^{(\gamma)} = 0.03\ \text{eV}$

Model Uncertainty in T_c: how does changing the cutoff temperature between the two regimes affect the activation energies?

$$T_c \in \{40,\ 45,\ 50,\ 55\}\,^\circ C$$

$$\sigma_{\text{sys}}^{(T_c)} = \frac{\max_i\ E_a(T_c^{(i)}) - \min_i\ E_a(T_c^{(i)})}{2}$$

T < T_c: $\sigma_{\text{sys}}^{(T_c)} = 0.008\,\text{eV}$ **T >= T_c:** $\sigma_{\text{sys}}^{(T_c)} = 0.8\,\text{eV}$

Final Results (eV):

$$E_a^{\text{low}} = -0.024 \pm 0.010\,(\text{stat.}) \pm 0.03\,(\text{sys.})$$

$$E_a^{\text{high}} = 2.4 \pm 0.8\,(\text{stat.}) \pm 0.8\,(\text{sys.})$$

Conclusions

$$E_a^{\mathrm{low}} = -0.024 \pm 0.010\,(\mathrm{stat.}) \pm 0.03\,(\mathrm{sys.})$$

$$E_a^{\mathrm{high}} = 2.4 \pm 0.8\,(\mathrm{stat.}) \pm 0.8\,(\mathrm{sys.})$$

Pagano et. al

Ea (T < 10) = 0.57

Ea (T> 10) = 1.18

- **Sharp increase in the net current R is observed;** unlikely to be explained by systematic environmental/apparatus effects
- **Fitting SRH model shows two plausible T regimes;** bad fit statistics (p-values) are likely due to systematic apparatus errors
- **Calculated activation energies deviate by 1.2 σ (high T) and 20 σ (low T) from Pagano et. Al**; not a direct comparison due to differing transition temperatures as well as different apparatus configurations (Over Voltage)
- **Overall:** good qualitative evidence for strong high-temperature dependence of dark current rate in a CW SiPM, but quantative agreement with SRH model, particularly at low T, remains to be shown

Suggestions for future experiments

- More data is necessary in the **temperature range T = 50 – 55 C**
  - For lower temperatures, physical explanation of why Ea is so small?
- **Plastic scintillator behavior is not considered;** other circuit elements that generate dark current? Differences between CW and Pagano et. Al study
- Remove apparatus error by **preventing CW from turning off**
- **Apply liquid nitrogen to the CW** to measure very low-T behavior of dark current
- Isolate dark current magnitude from total current

Thank You!

- **Lab partner:** Max Misterka
- **Jlab Instructional Staff:** Prof. Janet Conrad, Jixiang Yang, Simon Rothman, Vinh Tran, Atissa Banuazizi
- **JLab Technical Staff:** Dr. Sean Robinson, Dr. Aaron Pilarcik, Cory Romanov
- **Claude Code** was used to generate code for data analysis

Works Cited

- Source: Pagano et. Al, https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=6262467